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Major review

Acute macular neuroretinopathy: A comprehensive review of the literature



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ABSTRACT

Acute macular neuroretinopathy is a relatively rare condition originally defined by the presence of intraretinal, reddish-brown, wedge-shaped lesions, the apices of which tend to point toward the fovea. Acute onset of paracentral scotomas corresponding to the clinically evident lesions is both common and characteristic. Although the pathogenesis of acute macular neuroretinopathy is complex, recent research suggests a microvascular etiology. Advances in multimodal imaging have enabled better characterization of this retinal disorder and have led to newly proposed diagnostic criteria. We review 101 reported cases in the English and non-English language literature identified from 1975, when acute macular neuroretinopathy was first described, to December, 2014. We discuss common risk factors, demographic and clinical characteristics, and multimodal imaging findings, which together provide insights into pathogenesis and guide areas of future investigation.

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1. Introduction

Acute macular neuroretinopathy (AMN) is a rare retinal disorder initially described by Bos and Deutman in 1975.⁵ Characteristic lesions of AMN arise acutely in the macula and are classically described as reddish-brown and wedge-shaped, the apices of which tend to be directed toward the fovea often in a petaloid or tear-drop configuration. Acute, symptomatic photopsias and paracentral scotomas associated with mild loss of vision are common at the onset of this condition. Patients tend to be young, white, and female. Although a number of antecedent triggers have been associated with the condition, studies using spectral-domain optical coherence tomography (SD-OCT) have only recently identified ischemia involving the deep retinal capillary plexus as a possible pathogenic mechanism.⁵²

Turbeville and colleagues⁶⁹ published a major review of 41 cases of AMN in 2003, in which data regarding epidemiology, clinical characteristics, and fluorescein angiographic features were reviewed. The authors also analyzed their summary data to identify possible risk factors and to suggest future areas of research. Our comprehensive review of 156 eyes of 101 cases reported through December, 2014, more than doubles the number of total cases previously reviewed and analyzed. Particular emphasis has been placed on the results of more recently published studies using multimodal imaging techniques, many of which have provided important insights into the pathogenesis of AMN.

2. Results

2.1. Environmental triggers

Several associations or risk factors have been identified in patients with AMN, as summarized in Table 1. The most commonly reported associations are a nonspecific flu-like illness or fever (47.5%), use of oral contraceptives (35.6%),

and exposure to either epinephrine or ephedrine (7.9%). Antecedent trauma (5.9%) was also noted⁹ and included 4 cases of nonocular injury sustained by motor vehicle accident and 1 case of head trauma following a 3-meter fall. Several cases with systemic shock (5.0%) were also identified, including 1 case each of cardiac arrest, toxic shock syndrome, severe blood loss during total left hip arthroplasty, anaphylactic shock, and severe hypotension associated with epidural anesthesia. Other less common associations with 2 cases each (2.0%) included IV contrast exposure, pre-eclampsia, postpartum hypotension, and caffeine consumption—10 cups per day in 1 case and 2–3 cups per day in another.

2.2. Demographic characteristics

One hundred fifty-six eyes of 101 cases are summarized in Table 2.^{1–6,10,12–28,30–32,34–46,49–59,64,65,67,68,70,73} These clinical cases were reported from 13 countries, with roughly half (49.5%) originating from the USA. The vast majority (84.2%) of patients were female (M:F ratio = 0.16:1). The mean age of initial presentation was 29.5 years (median 26, range 12–65), with over half of the reported cases (51.5%) occurring in the third decade of life (Fig. 1). Although the youngest patient with AMN in the review by Turbeville and colleagues⁶⁹ was 17 years old, the present study identified 2 reports of AMN in children: a 12-year-old girl and a 15-year-old boy. Only 6 patients (6.0%) in were aged 60 years or older. Race was reported in only 35 of 101 patients (34.7%) and is summarized in Table 3. Among these 35 patients, 28 (80.0%) were non-Latino white, 3 (8.6%) were Asian-Indian, and 2 each (5.7%) were black and Latino.

2.3. Clinical characteristics

Acute macular neuroretinopathy was bilateral in 54.4% of patients. The vast majority (98.0%) with AMN reported visual symptoms, including scotomas in 73 cases (72.3%), decreased vision in 16 cases (15.8%), blurred vision in 12 cases (11.8%), floaters in 4 cases (4.0%), and metamorphopsia in 3 cases (3.0%). Visual acuity at presentation was generally good, documented to be 20/40 or better in 126 eyes (80.8%) and 20/200 or worse in 9 eyes (5.8%). At final follow-up, the timing of which varied considerably from case to case, persistent scotomas were present in 83 eyes (53.2%), visual acuity was 20/40 or better in 78 eyes (50.0%), and only 1 eye had a visual acuity of 20/200 or worse.

The distinctive wedge-shaped lesions (Fig. 2A–C) occurred in 38 eyes (24.4%). The configuration was described as petaloid in 14 eyes (9.0%), oval in 10 eyes (6.4%), tear-drop shaped in 6 eyes (3.8%), and horseshoe shaped in 1 eye (Fig. 2). The clinically observed lesions assumed a reddish-brown or orange color in 86 eyes (55.1%) but were reported to be hypopigmented or grayish-white in 3 eyes (1.9%). Nine eyes (5.8%) had no clinically identifiable changes, and near-infrared reflectance imaging was necessary to reveal the paracentral retinal defects. Superficial retinal hemorrhages (Fig. 2J and K) and macular edema were identified in 5 eyes (3.2%).^{21,55} Accompanying optic disc edema was a rare clinical feature.^{60,62,65} A single case of AMN occurred in association with multiple evanescent white dot syndrome.²¹

Table 1 – Disease characteristics

Finding	Number (% of total) ^a
Symptoms	
Scotomas	73 (72.3)
Decreased vision	16 (15.8)
Blurry vision	12 (11.8)
Laterality	
Bilateral	55 (54.4)
Unilateral	45 (44.5)
Not reported	1 (1.0)
Associated factors ^a	
Infection or febrile illness ^b	48 (47.5)
Oral contraceptives	36 (35.6)
Epinephrine/Ephedrine	8 (7.9)
Severe bodily/nonocular trauma	6 (5.9)
Systemic shock	5 (5.0)

^a Some cases have more than one associated factors.

^b This category includes description such as influenza, upper respiratory infection, sinusitis, enteritis, pharyngitis, and bronchitis.

Table 2 – Case summary

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Bos et al	1975	29	F	Sudden onset of “black dots” and scotoma in central field OU	OCP	Not reported	Dark, brown-reddish flecks in the superficial retina, triangular lesion pointed toward the fovea	Goldmann: normal peripheral boundaries, dense paracentral scotomas; Static Tubinger Perimetry: dense scotomas OU, slight decrease in central light sensitivity	Not reported	Not reported
Bos et al	1975	33	F	Sudden onset of scotoma OS	OCP	20/20 OS	A dark brown-reddish spot in the macula close to the fovea	Static Tubinger Perimetry: scotomas	Not reported	Not reported
Bos et al	1975	24	F	Disturbance of central vision	Enteritis, OCP	20/20 OU	Dark-red, wedge-shaped lesions pointing toward the fovea with a butterfly-shaped pattern in superficial retina OU	Static Tubinger Perimetry: paracentral scotoma OD; Amsler: 3 small scotomas close to fixation point OS	Not reported	Not reported
Bos et al	1975	32	F	Scotomas OU	Influenza, OCP	20/25 OD, 20/66 OS	Darkish red wedge-shaped lesions in superficial retina OU	Amsler: paracentral scotoma OD; large scotoma temporal to fixation OS	Not reported	Not reported
Bos et al	1975	30	F	Paracentral scotomas	Influenza, OCP	20/20 OD, 20/100 OS	Wedge-shaped lesions in the superficial retina OU	Paracentral scotomas	Not reported	Not reported
Bos et al	1975	23	M	Nonspecific vision loss	Influenza	20/25 OD	A dark dot in the superficial retina nasal to fovea OD	Small, dense scotoma corresponding to the lesion	Not reported	Not reported
Sanders et al	1976	22	F	Sudden central visual loss, pain with EOM	Not reported	6/60 OD	Minimal stippling of the macula	Central scotoma	Not reported	Not reported
Rush et al	1977	24	F	Three tear-shaped scotomas	Pharyngitis, OCP	20/20 OS	Three faint, discrete, reddish-brown wedge-shaped lesions pointing toward the fovea	Amsler and Goldmann: 3 wedge-shaped parafoveal scotomas pointing toward fixation	Not reported	Not reported
Neetens et al	1978	27	F	Not reported	Not reported	Not reported	Teardrop lesion	Amsler: wedge-shaped scotoma	Not reported	Not reported

Priluck et al	1978	17	M	Sudden onset of central blue-green shadows	Not reported	20/20 OU	Irregular dark appearing disturbance of the deep sensory retina near fovea, a small intraretinal hemorrhage inferior to left macula	Goldmann and Amsler: parafoveal scotoma OU	Not reported	Not reported
Guzak et al	1983	27	F	Sudden paracentral scotomas	Intravenous contrast (Renografin 76), Premarin cream	20/25 OD, 20/20 OS	Subtle gray opacification of retina in parafoveal retina with a central red spot, evolving in 10 days to wedge-shaped radially oriented lesions in the fovea	Goldmann & Amsler: paracentral scotoma OU	Not reported	Not reported
Guzak et al	1983	24	F	Sudden metamorphopsia	Infectious mononucleosis, OCP, intravenous contrast (Conray 43)	20/40 OU	Geographic pattern OD, wedge-shaped lesion oriented radially toward the fovea OS	Goldmann & Amsler: paracentral scotoma OU	Not reported	Not reported
Sieving et al	1984	20	F	Sudden loss of vision OD	OCP	20/40 OD	3 parafoveal burnt-orange wedge-shaped lesions in the nasal, parafoveal region, slight darkening of the foveola, subtle altered reflex superonasal to the parafoveal lesions	Amsler: 3 temporal paracentral scotomas corresponding to the parafoveal lesions; Goldmann: 1 wedge-shaped scotoma corresponding to the area with subtle, altered reflex	Not reported	Not reported
Van herck et al	1984	50	F	Sudden brown spot OS	Bronchitis	20/100 OS	Dark-red spot overlying the macula (2 disc diameter in size)	Amsler & Friedmann: dense paracentral scotoma	Not reported	Not reported
Van Herck et al	1984	23	F	Difficulty reading OS	Influenza	20/20 OS	Small, dark-red, wedge-shaped spot nasal to the fovea	Amsler & Friedmann: small dense paracentral scotoma temporal to the fixation	Not reported	Not reported
Van Herck et al	1984	47	F	“Spot” OS	Influenza	20/20 OS	Dark-red, wedge-shaped lesion nasal above the fovea	Amsler: paracentral wedge-shaped scotoma temporal inferior to fixation	Not reported	Not reported

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Table 2 – (continued)

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Van Herck et al	1984	24	F	Sudden onset of gray spots	Rovamycine, Influenza, OCP	6/10 OD, 9/10 OS	3 weeks: increased reflex in the posterior pole 2 months: parafoveal wedge-shaped lesions	Amsler: paracentral scotomas OU	Not reported	Not reported
Rait et al	1987	21	F	Sudden onset of spots in central vision	URI, OCP	20/20 OU	Red-brown scalloped areas surrounding the fovea	Amsler & Goldmann: incongruous paracentral scotomas OU	Not reported	Not reported
Gass et al	1989	22	F	Scotomas in central visual field	OCP	20/30 OS	Unilateral; Multiple, variable-sized, round and oval, red-orange lesions in outer retina in the central macula	Amsler: multiple round paracentral scotomas	Not reported	Not reported
Gass et al	1989	30	F	Blurred vision, large temporal scotoma, shimmering photopsias	Not reported	20/20	Unilateral; multiple peripheral macular and juxtapapillary small red-orange lesions at the level of the RPE	Goldmann: enlarged blind spot and multiple paracentral scotomas	Not reported	Not reported
Miller et al	1989	26	F	Gray spots	Influenza, OCP	6/5 OU	Irregular dark areas around the fovea OU	Amsler, Bjerrum & Goldmann: parafoveal scotoma	Not reported	Not reported
Miller et al	1989	23	F	Progressive gray scotomas	Not reported	6/5 OU	Dark lesions in the macula OU	Amsler: parafoveal scotoma	Not reported	Not reported
Miller et al	1989	23	F	Sudden onset of spots	OCP, migraine	6/6 OU	Not reported	Amsler & Tübingen: paracentral scotomas OU	Not reported	Not reported
Miller et al	1989	25	F	Patches of blurred vision	Influenza, OCP	6/9 OD, 6/6 OS	Several reddish-brown lesions in macula OU	Goldmann & Friedmann: normal OD, relative scotoma OS	Not reported	Not reported
Miller et al	1989	27	F	Sudden onset of spots OU	Influenza, OCP	6/6 OU	Parafoveal reddish areas OU	Amsler: paracentral scotoma OU	Not reported	Not reported
O'Brien et al	1989	28	F	Sudden onset of central vision loss	IV ephedrine	20/200 OU	Bilateral; Central red spots evolving to red wedge-shaped lesions in outer retina in the fovea OU	Amsler: paracentral scotomas OU	Not reported	Not reported
O'Brien et al	1989	26	F	Sudden onset of vision loss, paracentral scotoma	IV epinephrine	20/30 OU	Deep red outer retinal lesion in the macula OU	Amsler: paracentral scotomas OU	Not reported	Not reported

O'Brien et al	1989	23	F	Sudden onset of paracentral scotoma OU	IV epinephrine	Not reported	Deep red macular lesions OS > OD	Not reported	Not reported	Not reported
Leys et al	1991	22	F	Sudden onset of paracentral vision loss OD	Anaphylactic shock, epinephrine, OCP	20/20 OD	Not reported	Dense central scotoma OD	Not reported	Not reported
Leys et al	1991	26	F	Sudden onset of a single dark spot	Postpartum hypotension, Epinephrine	20/20 OD, 20/25 OS	Small lesions mimicking hemorrhage in the papillomacular bundle OD, irregular shaped dark-red area OS	Goldmann: scotoma superior to fixation OS	Not reported	Not reported
Singh et al	1991	24	F	Gradual onset of inferior temporal scotoma	Not reported	20/25 OS	Multiple, variable size, red-orange lesions at the level of RPE in macula OS	Goldmann: 20° absolute scotoma encompassing the physiologic blind spot OS	Not reported	Not reported
Desai et al	1993	18	M	Subacute development of paracentral scotomas	IV epinephrine	6/9 OU	Reddish-brown petalloid lesion at the level of outer neurosensory retina OU	Amsler grid: paracentral scotomas OU	Not reported	Not reported
Gillies et al	1997	45	M	Sudden onset of scotoma in central field OS	Trauma (nonocular)	6/12 OS	Several small foveal hemorrhages evolving in 3 weeks to several dark oval patches around the fovea	Amsler: scotoma inferotemporal to fixation OS	Not reported	Not reported
Gillies et al	1997	17	M	Sudden onset of spot to the left of fixation OS	Trauma (nonocular)	6/5 OS	Not reported	Amsler: oval-shaped scotoma temporal to fixation OS	Not reported	Not reported
Gillies et al	1997	33	M	Black spot in central field	Trauma (nonocular)	4 days: 6/60 OD; 10 days: 6/24 OD; 6 weeks: 6/9 OD	Macular hemorrhages and edema, evolving in 6 weeks to numerous dark patches (honeycomb effect at the fovea) OD	Not reported	Not reported	Not reported
Gillies et al	1997	53	M	Dark patches in field OU	Trauma (nonocular)	6/12 OD, 6/9 OS	1.5 DD well-circumscribed curvilinear dark patch involving the fovea OU	Dense central scotomas OU	Not reported	Not reported
Amin et al	1998	40	F	Sudden onset of swirling central scotoma	Flu-like illness, OCP	20/20 OU	Bilateral reddish-brown areas in the macula OU	HVF 30-2: paracentral scotomas OD > OS (denser temporally)	Not reported	Not reported

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Table 2 – (continued)

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Tingsgaard et al	1999	21	F	Paracentral scotomas OU	Flu-like illness	0.05 OU	Abnormal foveal reflex (no distinctly visible fundus lesions) OU	Amsler: small semilunar paracentral scotoma OU	Not reported	Not reported
Feigl et al	2000	33	F	Sudden onset of central, gray, swirling scotomas OD	Viral illness	20/20 OD	Red-brown lesions in the macula OD	Amsler: metamorphopsia in the superior temporal quadrant OD; HVF 10-2: paracentral scotomas	Not reported	Band of hyper-reflectivity (115 μ m) overlying an intact band corresponding to RPE/choriocapillaris complex, sparing the fovea
Kerrison et al	2000	25	F	Sudden onset of painless loss of vision OU	Hypotension during epidural anesthesia	20/20 OU	Reddish, petaloid lesions OU	Central defects OU	Not reported	Not reported
Gandorfer et al	2002	23	F	Sudden onset of central scotoma OU	Flu-like illness, OCP	20/20 OU	Reddish-brown wedge-shaped areas in the macula OU	Not reported	Dark, well-defined areas in the macula	Not reported
Gomez-Torreiro et al	2002	27	F	Sudden onset of black wedge-shaped scotoma	Not reported	20/20 OS	Poorly defined, reddish-orange area nasal to macula OS	Amsler: temporal paracentral scotoma OS; Microperimetry on SLO: absolute scotoma close to fovea, relative scotoma in the rest of the lesion	Well-defined triangular lesion nasal in location and pointing to the fovea	Not reported
Douglas et al	2003	21	F	Sudden onset of floaters, photophobia, and “images” OU	Febrile illness, OCP	6/6 OD, 6/6 OS	Circular lesion OD, wedge-shaped lesions pointing toward the fovea OS (base lighter in color with less defined boundaries)	Medmont automated perimeter: normal OD, small paracentral scotoma OS	Not reported	Not reported
Maturi et al	2003	12	F	Sudden onset of localized field loss OS	Viral illness	20/20 OS	Unilateral; slightly dark reddish-brown deep macular lesion OS	Goldmann: dense, nasal perifoveal scotoma; steeply marginated	Not reported	Not reported
Browning et al	2003	23	F	Central nonspecific visual disturbance	Flu-like illness, OCP	20/20 OD, 20/17 OS	Red wedge-shaped lesions centered around the fovea OU	HVF: scotoma OU	Well-demarcated wedge-shaped lesions	Not reported
Shah et al	2003	50	F	Black spots OS	Not reported	20/20 OS	Scattered orange petaloid lesions in the foveola OS	Enlarged blind spot	Not reported	Not reported

Watzke et al	2004	61	F	Distortion, ring-shaped scotoma OS	Not reported	20/15 OS	Unilateral; circular zone of slightly pigmented and flat outer retina and RPE within 500 µm of the foveola OS	Amsler: annular dark scotoma around fixation	Not reported	3-year follow-up: defects in the outer high signal band
Shukla et al	2005	30	M	Gradual onset of blurry vision OS × 6 months	Not reported	6/9 OS	Subtle red wedge-shaped lesion above the fovea pointing toward the fovea OS	Amsler: scotoma corresponding to the macular lesion OS	Not reported	Distinct retinal thinning in the area of the lesion, more prominent thinning of the inner retinal layers
Chan et al	2005	34	F	Sudden onset of blurred vision OS	Flu-like illness	6/6 OS	Dark reddish-brown oval lesions at fovea OS	HVF 24-2: dense paracentral scotoma OS	Not reported	Acute period: A band of hyper-reflectivity between spared foveal depression and underlying intact RPE-choriocapillaris; 3-month follow-up: localized loss of the hyper-reflective layer overlying the RPE
Mirshahi et al	2006	22	M	Subacute onset of paracentral scotoma × 6 days	Not reported	6/6 OU	A sharply delineated reddish-brown area superior to the foveola in the outer retinal segments OD, 3 small paracentral lesions inferonasal to the foveola OS	Amsler: paracentral scotoma, Octopus 10-2: paracentral dense scotoma	Not reported	Not reported
Salom et al	2007	22	F	Sudden onset of central scotomas and vision loss	Flu-like illness, OCP	1.0 OD, 1.0 OS	Parafoveal reddish-brown, round spots with poorly defined borders OU	Deep central scotomas	Not reported	No abnormality detected
Monson et al	2007	51	F	Sudden onset of gray, oval-shaped paracentral scotoma OD	Not reported	20/25 OU	Unilateral; focal, reddish petaloid lesion superior to fixation OD	Amsler: paracentral scotoma	Not reported	Stratus OCT: normal, ultrahigh-resolution OCT: focal depression of the external limiting membrane, IS/OS junction, photoreceptors, and RPE; changes in the photoreceptor layer OS in the region of petaloid lesions

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Table 2 – (continued)

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Corver et al	2007	30	F	Sudden onset of black spots in central field, flickering lights OU	Postpartum hypotension, methyl-ergometrin and etilefrine (sympathomimetic)	20/20 OU	Bilateral; reddish-brown petaloid lesions	Amsler: well-delineated, U-shaped scotoma OU, HVF 10-2: U-shaped paracentral scotoma OU	Not reported	(normal inner retina); 3-month follow-up: realignment of the outer photoreceptor layer and the IS/OS junction 16-month follow-up: retinal thinning corresponding to the lesions
Corver et al	2007	19	F	Sudden onset of central scotoma OU	Influenza	Not reported	Bilateral; horse-shoe shaped macular lesions	Paracentral scotoma OU	Not reported	Not reported
El-Dairi et al	2009	21	F	Sudden onset of blurred vision and black spots	Epinephrine, OCP	20/40 OD, 20/30 OS	Reddish-brown petal-shaped lesions OU	HVF: paracentral scotoma OU	Not reported	Acute period: abnormal increased signal at the OS/IS junction, with focal area of relative thinning; 6-week follow-up: increased signal disappeared and replaced by focal atrophy located at the photoreceptor layer, progressed thinning of the retina
Hughes et al	2009	27	F	Sudden onset of paracentral scotoma OU	Flu-like illness	6/9 OU	Subtle patchy red discoloration OU	Amsler: paracentral scotoma	SLO infrared imaging: dark areas corresponding to abnormalities on Amsler testing	High-resolution OCT: disruption of the IS/OS junction, 8-month follow-up: partial reconstitution of the IS/OS junction with associated focal thinning of the outer nuclear layer
Engelbert et al	2009	31	F	Sudden onset of paracentral scotoma	Flu-like illness	20/25 OU	Reddish-brown, wedge-shaped lesions pointing toward the fovea	Amsler: paracentral scotoma	Not reported	SD-OCT: disruption of the IS/OS junction, thinning of the outer nuclear layer

Neuhann et al	2010	24	F	Sudden onset of scotomas	Flu-like illness, OCP	20/20 OU	Two sharply demarcated oval/petal-shaped parafoveal lesions OU	Amsler: 2 paracentral scotomas	Not reported	indicating photoreceptor loss, 5-month follow-up: persistent photoreceptor loss SD-OCT: focal discontinuation of the hyper-reflective bands of the outer retina OU
Neuhann et al	2010	21	F	Sudden onset of scotomas	Flu-like illness	20/20 OU	Two sharply demarcated oval/petal-shaped parafoveal lesions OU	Amsler: 2 paracentral scotomas	Not reported	SD-OCT: focal discontinuation of the signal of the outer retinal structures
Grover et al	2011	25	F	Sudden onset of blurred vision	URI	20/20 OU	Pigmentary changes in the papillomacular bundle, wedge-shaped lesions with apices oriented toward the fovea OU	Amsler: bitemporal paracentral scotomas; HVF 30-2: paracentral scotomas OU, microperimetry (MP-1): focal elevation in threshold correlating with the wedge-shaped defects	Wedge-shaped lesions with their apices oriented toward the fovea	SD-OCT: changes at the IS-OS junction with thickened OPL overlying these areas
Rifkin et al	2011	16	F	C-shaped scotomas OU	Flu-like illness, OCP	20/20 OU	Reddish-brown, petaloid lesions OU	Amsler: C-shaped scotomas; HVF: cecocentral scotomas extending superotemporally OU	Not reported	SD-OCT: discontinuity of the outer retina, specifically the IS/OS junction
Munch et al	2011	25	F	Sudden onset of blurred vision and dark spots in central field OU	IV cytokine-releasing immune-therapy, OCP	0.3 OD, 0.8 OS	Sharply demarcated dark lesions OU	Large paracentral relative scotoma OU	Sharply demarcated dark lesions	Acute period: hyper-reflectivity of the outer nuclear layer and hyporeflexivity of the photoreceptor/RPE complex; 5 month follow-up: previously described abnormalities have largely resolved
Matsuo et al	2011	48	F	Sudden onset of small paracentral scotoma OD	OCP	1.2 OD	Small reddish-brown lesion, parafoveal in location OD	Amsler: inferior paracentral scotoma (not detected on HVF 10-2)	Not reported	No abnormality detected
Matsuo et al	2011	39	M		Not reported	1.2 OS		Not reported	Not reported	Stratus OCT: hyper-reflectivity over the

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Table 2 – (continued)

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Yeh et al	2011	24	F	Sudden onset of paracentral scotoma OS Central scotoma OU	Not reported	20/25 OD, 20/20 OS	Small reddish-brown lesion, parafoveal in location OS Subtle reddish boot-shaped lesion OD, wedge-shaped lesion OS	Amsler: well-defined boot-shaped scotoma OD and wedge-shaped scotoma OS	Well-defined boot-shaped lesion OD, wedge-shaped lesion OS	intact RPE; 2-month follow-up: hyper-reflectivity faded SD-OCT: attenuation of outer nuclear layer and photoreceptor inner segment-outer segment junction no changes after 4 months
Yeh et al	2011	17	F	Two shadows associated with photopsias	Flu-like illness	20/20 OS	Abnormal foveal reflex OS	Amsler: 2 areas of visual blur temporal to fovea OS	2 discrete lesions nasal to fovea	SD-OCT: attenuation of the IS/OS junction, focal thinning of the outer nuclear layer
Vance et al	2011	34	F	Dark spot OS	Not reported	20/25 OS	Small dark area superior to the fovea OS	Defect inferior to fixation	Not reported	SD-OCT: focal area of IS/OS boundary loss, 1-month follow-up: no change in IS/OS boundary loss, thinning of the outer nuclear layer
Vance et al	2011	28	F	Two black spots inferior to fixation OD	Sinus infection	20/20 OD	2 small dark spots superior to the fovea OD	No defect by testing	Not reported	SD-OCT: focal loss of the IS/OS boundary, subtle ONL thinning over the area of IS/OS defect, 1-month follow-up: IS/OS boundary was re-established, persistent ONL thinning
Vance et al	2011	31	F	Not reported	Flu-like illness	20/25 OU	Reddish-brown, wedge-shaped lesions OU	Not reported	Not reported	SD-OCT: IS/OS defects, thinner overlying ONL OU; 5-month follow-up: persistent IS/OS boundary loss, no progression of ONL thinning
Vance et al	2011	34	F	Sudden onset of scotomas OU	Hypertension during pregnancy	20/20 OU	Reddish-brown, wedge-shaped lesions OU	Not reported	Wedge-shaped lesions	SD-OCT: focal areas of IS/OS boundary loss, ONL thinning; 1-month follow-up:

Maschi et al	2011	17	F	Sudden onset of nonspecific visual disturbance OU	Flu-like illness, OCP	10/10 OU	Wedge-shaped lesion OU	HVF: small bilateral paracentral scotomas OU	Hyporeflective areas corresponding to scotomas	return of some of the IS/OS boundary, persistent ONL thinning TD-OCT: no lesion, SD-OCT: hyper-reflective lesion located in the outer nuclear layer at the level of ELM, broadening of the IS/OS junction line; 10-day follow-up: hyper-reflective lesion disappeared, IS/OS junction became hyporeflective 3-week follow-up: SD-OCT: hyporeflective lesions between the photoreceptor layer and the ELM; 6-month follow-up: no change SD-OCT: hyper-reflective OCT backscatter at the outer plexiform layer; 1-week follow-up: abnormal reflectivity of the IS/OS junction, hyper-reflectivity in the ONL, decrease in outer retinal thickness, OPL no longer hyper-reflective
Maschi et al	2011	25	F	Sudden onset of sparkling scotoma in central field	Flu-like illness, OCP	10/10 OU	Petalloid lesions OS	Bilateral; HVF: paracentral scotoma OS > OD	Not reported	SD-OCT: focal highly reflective band of the outer plexiform layer extending into the outer nuclear layer, with a slight hyporeflectivity of the ELM; 3-day
Baumuller et al	2012	29	F	Nonspecific vision loss OD	Not reported	Not reported	Wedge-shaped dark reddish lesion OD	Central scotoma OD	Not reported	(continued on next page)
Azar et al	2012	32	F	Small dark spot superior to fixation OS	Flu-like illness	20/20	No lesions observed OS	Amsler: paracentral scotoma	Hyporeflective, sharply defined oval lesions OS	

Table 2 – (continued)

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Ranjan et al	2012	26	F	Black spots, scotomas OU	Postpartum	20/20 OU	Oval-shaped lesion OD, well-circumscribed bean-shaped area of retinal discoloration OS	Amsler: central scotoma OU; Automated: dense scotomas OU	Not reported	follow-up: highly reflective band of the ONL with disruption of the IS/OS interface, OPL normalized, disruption of the ELM/RPE inner band; 4-month follow-up: focal depression of IS/OS junction with a wider disruption of the RPE inner band and thinning of the outer nuclear layer SD-OCT: well-delineated defect in the reflectivity of outer retinal layer adjacent to RPE
Fawzi et al	2012	41	F	Sudden onset of vision loss OU	Flu-like illness	20/25 OD, 7/200 OS	Dark, wedge-shaped lesion OU	Automated HVF: paracentral scotoma corresponding to lesion OD	Well-defined lesions	TD-ODCT: hyper-reflectivity of the ONL with obscuration of the underlying outer retina OU; SD-ODCT: ONL hyper-reflectivity, disruption of IS/OS, OS/RPE junctions; 14 follow-up: thinning of ONL, slight attenuation of IS/OS junction, and persistent absence of OS/RPE junction SD-OCT: thinning of ONL and outer segment line, disruption of IS/OS and OS/RPE junction SD-OCT: decreased signal of IS/OS and
Fawzi et al	2012	45	F	Gradual onset of iron-shaped blind spot × 2 months OD	Not reported	20/20 OD	Foveal granularity OD, red-brown oval lesion OD	Amsler: triangular scotoma OD; HVF: dense paracentral scotoma OD	Hyporeflective lesions	
Fawzi et al	2012	45	F	Sudden onset of vision loss OS	Motor vehicle accident with severe	CF @ 3-ft OS		Not reported		

Fawzi et al	2012	27	F	Gradual onset of multiple central scotomas, loss of vision OS × 3 months	acceleration/ deceleration forces (no physical injuries) Not reported	20/60 OS	Brownish red petalloid lesion in the fovea OS No lesions observed	Not reported	Hyporeflective petalloid lesions OS Multiple round, dark lesions in the fovea	OS/RPE. Decreased OS/RPE correlated with IR images SD-OCT: no abnormality detected
Fawzi et al	2012	53	F	Sudden onset of iron-shaped scotoma OD	Not reported	20/20 OD	Paracentral, triangular lesions OD	Not reported	Hyporeflective, triangular lesion	SD-OCT: hyper-reflectivity of OPL/ ONL, normal IS/OS junction, disruption of RPE/OS junction; 2-week follow-up: resolution of the hyper-reflectivity at OPL, disruption of IS/ OS junction, and OS/ RPE; 1-month follow-up: increased OS/RPE disruption (normal IS/OS junction); 6 and 8 weeks: incomplete normalization of the IS/OS junction with persistent OS/RPE absence
Fawzi et al	2012	42	F	Gradual onset of central scotoma OD × 6 months	Not reported	20/20 OD	Not reported	Not reported	Hyporeflective wedge-shaped lesion	6 months after onset of symptoms: SD-OCT: ONL thinning, OS/RPE disruption, relative preservation of IS/OS junction; 10-month follow-up: persistent ONL thinning, OS/RPE junction disruption with outer segment thinning
Fawzi et al	2012	26	F	Central wedge-shaped gray-black scotomas OU	Viral illness	20/20 OD, 20/25 OS	Not reported	Not reported	Faint, round, hyporeflective lesions	SD-OCT: hyper-reflectivity of the OPL and outer segment involvement; 3-month follow-up: normal OPL, thinning of ONL,

(continued on next page)

Table 2 – (continued)

Author	Date	Age	Sex	Onset, laterality and quality of symptoms	Associations	VA	Lesion characteristics	Visual field testing	Infrared imaging	OCT characteristics
Fawzi et al	2012	27	F	Focal scotomas OU	Febrile illness, OCP	20/20 OU	Multiple, reddish teardrop-shaped lesions OU	Amsler: scotomas	Hyporeflective lesions	normal IS/OS junction, persistent OS/RPE junction disruption SD-OCT: loss of OS/RPE junction, thinning of IS/OS junction and ONL
Hashimoto et al	2012	41	F	Sudden onset of vision loss OS	OCP	20/13 OU	Bilateral; wedge-shaped, dark, reddish-brown lesion in the macula OU	Amsler, HVF 10-2: central scotoma	Hyporeflective lesions	TD-OCT: initial: disruption of the IS/OS junction with thinning of ONL; 45-month follow-up: recovery of the IS-OS junction
Gelman et al	2012	52	F	Sudden onset of scotomas and blurry vision OU	Not reported	20/100 OD, CF OS	Dark, wedge-shaped lesions OU	Central scotomas OU	Wedge-shaped, well-defined macular lesions	SD-OCT: loss of IS/OS junction OU, faint reflective band above IS/OS junction; 2-week follow-up: no change; 1-month follow-up: persistent IS/OS junction defects
Hansen et al	2013	26	F	Gradual onset of paracentral scotoma OS × 1 month	Viral prodrome	20/20 OS	Unilateral; dull foveal reflex OS	Amsler: 2 small scotomas temporal to fixation	Not reported	Loss of IS-OS and OS-RPE junctions OS
Sarraf et al	2013	61	M	“Gray spot” above fixation OD × 2 weeks	Stress, increased caffeine intake	20/25 OD	Normal foveal reflex, no lesions	Amsler: small paracentral scotoma superior to fixation	Subtle, wedge-like, dark gray lesion at the inferior fovea	Hyper-reflective plaque-like band at the junction of the OPL and INL extending into the INL
Sarraf et al	2013	65	F	“Purple” paracentral scotoma OS × 1 week	No associations	20/20 OS	Normal foveal reflex, no lesions	Not reported	Dark, well-demarcated, nummular lesion at the nasal fovea	Hyper-reflective band at the junction of the OPL and INL extending into the INL
Sarraf et al	2013	56	M	Paracentral scotomas OS > OD × 6 weeks	Post operative (total left hip arthroplasty) severe blood loss	20/15 OD, 20/25 OS	Normal foveal reflex, no lesions	Not reported	Dark macular lesions in a cloverleaf pattern, OS > OD	Perifoveal hyper-reflectivity of the OPL and INL with normal IS/OS and OS/RPE bands

Sarraf et al	2013	60	M	Paracentral scotoma OD × 2 days	Flu-like illness	20/20 OU	Ill-defined intraretinal white lesion OD	Not reported	Lesion appeared dark and more precisely delineated, OS > OD	Hyper-reflective plaque at the junction of the OPL and INL extending into the INL
Sarraf et al	2013	54	M	Inferonasal paracentral scotoma OS × 2 days	No associations	20/20 OU	Paracentral superotemporal intraretinal white lesion associated with a subtle intraretinal hemorrhage OS	Not reported	Lesion appeared dark with its borders more sharply outlined	Hyper-reflective plaque at the junction of the OPL and INL extending into the INL
Sarraf et al	2013	33	F	Central “spot” OS × 5 days	Remote history of toxic shock syndrome	20/20 OU	Pale, semitranslucent, teardrop-shaped lesion in the parafoveal region OS associated with an intraretinal hemorrhage	Not reported	Dark teardrop lesion, better outlined by NIR	Hyper-reflective band at the level of the OPL and ONL with attenuation of the underlying IS/OS and OS/RPE layers
Sarraf et al	2013	43	F	Paracentral scotoma, “claw-shaped gray area” just temporal to fixation OD	No associations	20/20 OD	Paracentral, well- demarcated, wedge- shaped opacification nasal to fixation OD	Not reported	Dark lesion, better visualized with NIR	Hyper-reflective band involving the OPL and ONL layers with associated disruption of the IS/ OS and OS/RPE layers
Sarraf et al	2013	35	F	“Blurry vision” in both eyes, OS > OD	Treatment of pre- eclampsia, symptoms developed during inpatient hospitalization	20/25 OD and 20/ 30 OS	Bilateral cotton-wool spots, and reddish- brown discoloration of the central macula OS > OD	Central scotoma OS	Bilateral, well- demarcated, petaloid-shaped, dark gray macular lesions	Involvement of the HFL and ONL, patchy dropout of the IS/OS and OS/RPE lines OD, pronounced attenuation of both layers OS
Sarraf et al	2013	21	F	Central “bright spot” OS × 18 days	No associations	20/25 OS	Barely detectable dark paracentral lesion OS	Not reported	Small, wedge- shaped, dark gray lesion involving the superotemporal fovea OS	Subtle hyper- reflective lesion involving the OPL and ONL layers with associated focal disruption of the IS/ OS and OS/RPE layers
Tsui et al	2013	32	F	Wedge-shaped paracentral scotoma OS	Caffeine	20/80 OS	Parafoveal wedge- shaped lesion	Not reported	Dark gray, wedge-shaped paracentral lesion	Band-like hyper- reflective lesion at the level of the inner nuclear layer (INL) extending from the outer plexiform layer (OPL) to the inner plexiform layer (continued on next page)

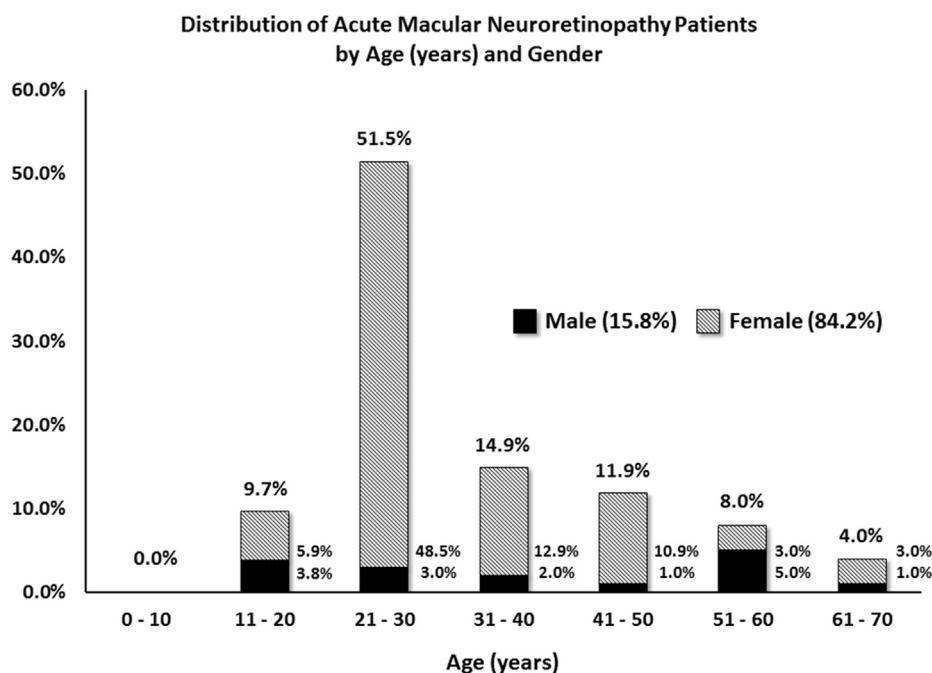


Fig. 1 – Histogram distribution of acute macular neuroretinopathy patients by age and gender. The mean age of initial presentation was 29.5 years (median 26, range 12–65).

2.4. Diagnostic testing and multimodal imaging

2.4.1. Near-infrared reflectance

The use of near-infrared reflectance (NIR) was reported in 50 eyes (32.1%), and in each instance, AMN lesions were reported to be dark or gray with well-demarcated margins (Fig. 3). In 2 eyes (4.0%) imaged with NIR, the edge or rim of the lesion appeared darker than the center. Among the eyes with discrete NIR lesions, definite clinical changes corresponding to the lesions were observed in 32 eyes (64.0%), subtle clinical changes in 9 eyes (18.0%), and no clinical changes in 9 eyes (18.0%). Among the 41 eyes with AMN lesions examined both

clinically and with NIR, the lesions tended to be defined more clearly using NIR (53.7%). NIR lesions were noted to correlate with scotomata in 7 eyes (14.0%).

2.4.2. Visual field

Most eyes (115 eyes, 73.7%) with AMN experienced one or more paracentral scotomas by Amsler grid, Goldmann, or Humphrey (10-2, 24-2, 30-2) visual field testing. Of these tested eyes, the shape of the visual field defect was not reported in 84 eyes (73.0%); however, for the 31 eyes (27%) for which detailed information was available, visual field abnormalities corresponded closely to the shape and location of the clinical lesion. The most common visual field defect detected was wedge-shaped (9 eyes, 29.0%), followed by U-shaped found in 4 eyes (13.0%), ring-shaped in 3 eyes (9.7%), round-shaped in 2 eyes (6.5%), semilunar shaped in 2 eyes (6.5%), oval shaped in 2 eyes (6.5%), and boot-shaped in 1 eye (3.3%). An enlarged blind spot was noted in 2 eyes (1.3%). The visual field was normal in only 2 eyes (1.3%). At final follow-up, persistent scotomas were noted in 83 eyes (53.2%). Sieving and colleagues⁶⁴ reported a patient with visual field changes that persisted for 9 years.

2.4.3. Fluorescein and indocyanine green angiography

The use of fluorescein angiography (FA) was reported in 97 eyes (62.2%). Normal angiograms were documented in the vast majority (72 eyes, 74.2%). Subtle hypofluorescence of the lesions was the most frequently described abnormality (20 eyes, 20.6%), and in 11 of these eyes (55.0%), the hypofluorescence was evident both early and late in the study. In 2 eyes (10.0%), hypofluorescence of the lesions was observed only in the late phase of the angiogram. Dilation of perimacular capillaries was seen in 3 eyes (3.1%). Punctate, early

Table 3 – Demographics for patients diagnosed with AMN

Country	Number (% of total)
Age at presentation, years	
Mean	29.5
Median	26
Range	12.0–65.0
Sex	
Male	16 (15.8)
Female	85 (84.2)
Ethnicity	
White	28 (27.7)
Asian-Indian	3 (3.0)
Black	2 (2.0)
Hispanic	2 (2.0)
NR	66 (65.3)
AMN, acute macular neuroretinopathy; NR, not reported. Data are no. (%) of patients unless stated otherwise.	

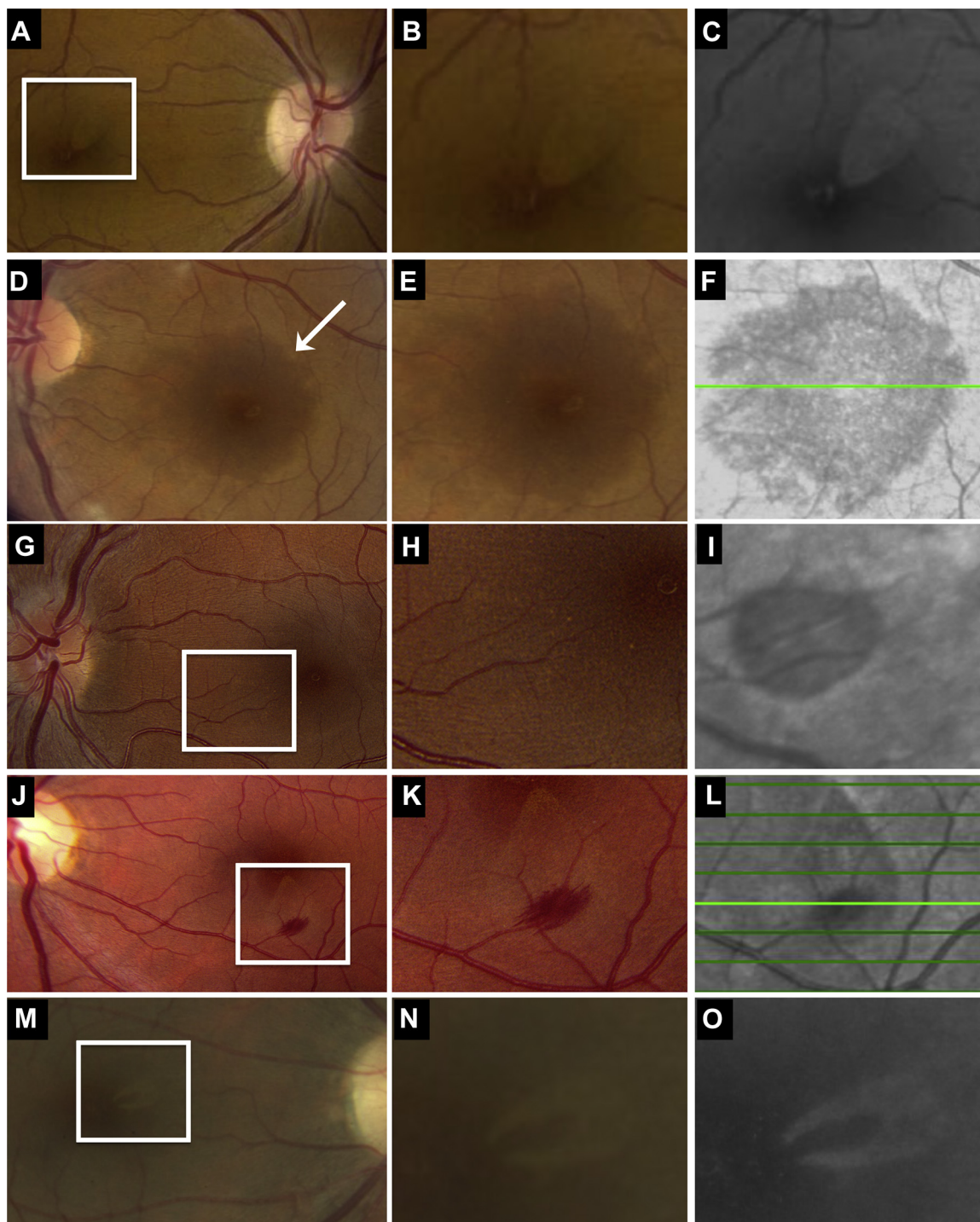


Fig. 2 – Acute macular neuroretinopathy lesion configurations by color fundus photographic, red-free, and near-infrared imaging: A–C: wedge, D–F: petalloid, G–I: oval, J–L: teardrop, and M–O: horse-shoe presentations. C, O: Ancillary red-free or F, I, L: near-infrared reflectance better delineate the lesion extent. Panels A–C reproduced from Fawzi et al¹² with permission from Wolters Kluwer Health. Panels D–F and J–L reproduced from Sarraf et al⁴⁴ with permission from American Medical Association.

hyperfluorescence and late staining of AMN lesions was present in 1 eye (1.0%). Kerrison and colleagues³⁴ described a case in which choroidal filling defects were observed, but otherwise choroidal vascular abnormalities were not noted on FA.

Indocyanine green angiography (ICGA) was used in 12 eyes (7.7%). Similar to FA, ICGA was typically normal in patients with AMN (10 eyes, 83.3%). Hashimoto and colleagues²⁹ observed early bilateral hypofluorescence with ICGA in a patient with AMN.

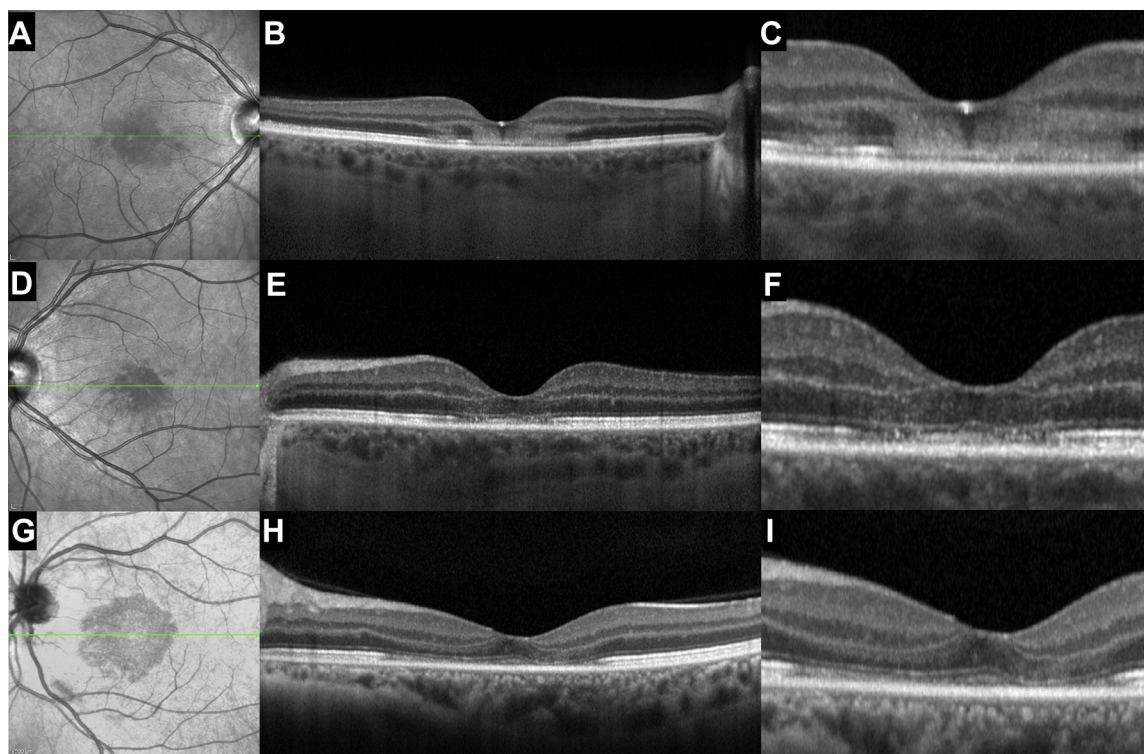


Fig. 3 – A, D, G: Near-infrared reflectance imaging from 3 individuals with acute macular neuroretinopathy demonstrates well-demarcated, dark (hyporeflective) macular lesions. B, C, E, F, H, I: Corresponding spectral-domain optical coherence tomography imaging from these 3 cases depicts hyper-reflectivity of the outer plexiform layer and adjacent outer nuclear layer with associated disruption of the inner segment ellipsoid.

2.4.4. Fundus autofluorescence

Fundus autofluorescence (FAF; Spectralis cSLO 488/521 nm) was observed in 8 of 156 eyes (5.1%) from 4 reports of AMN. The FAF was unremarkable in 4 of these imaged eyes (50.0%).^{26,49} Yeh and colleagues⁷⁴ described subtle areas of mild hypoautofluorescence in the region of clinically evident lesions in 2 eyes (28.6%). In this case, the FAF changes were less prominent than either infrared or SD-OCT abnormalities. Gelman and colleagues²² described bilateral focal increases in FAF associated with sudden development of scotomas in a patient with pancreatitis who presented with 5 days of blurred vision. They noted that the focal hyperautofluorescence correlated to regions of ellipsoid zone loss observed on SD-OCT. In addition, Fawzi and colleagues¹⁸ described relative hyperautofluorescence of clinical lesions on near-infrared FAF (NIA) and proposed that alterations in retinal pigment epithelium (RPE) melanin could be responsible for this observation.

2.4.5. Optical coherence tomography

Optical coherence tomography was reported in 97 of 156 eyes (62.1%), including 19 eyes (12.2%) that underwent time-domain optical coherence tomography and 78 eyes (50.0%) that underwent SD-OCT. Among the 19 eyes that underwent time-domain optical coherence tomography, 6 eyes (31.6%) showed no abnormalities^{40,44,48} and 13 eyes (68.4%) showed one or more abnormal findings including outer retinal

hyper-reflectivity (6 eyes, 31.6%), ellipsoid zone loss (3 eyes, 15.8%), generalized retinal thinning (2 eyes, 10.5%), and thinning of the outer nuclear layer (ONL) (2 eyes, 10.5%).

Among the 78 eyes that underwent SD-OCT, 1 eye (1.3%) showed no abnormalities,¹⁴ and 77 eyes (98.7%) showed one or more abnormal findings, including ellipsoid zone disruption in 37 eyes (47.4%), hyper-reflectivity of the ONL in 25 eyes (32.1%), thinning of the ONL in 14 eyes (17.9%), hyper-reflectivity of the outer plexiform layer (OPL) in 4 eyes (5.1%), hyper-reflectivity of Henle layer in 3 eyes (3.8%), hyporeflectivity of the photoreceptor/RPE complex in 3 eyes (3.8%), and generalized outer retinal loss in 1 eye (Figs. 3 and 4).^{2,4,7,18,26,39,43,44,49,61,63,67,71,74}

Time from initial to last time-domain optical coherence tomography varied from 6 to 180 weeks (mean 63.3; median 18.0). Persistent changes identified on time-domain optical coherence tomography at last visit included generalized retinal thinning observed in 5 eyes (26.3%), loss of hyper-reflectivity in 2 eyes (10.5%), and recovery of the ellipsoid zone in 2 eyes with initial ellipsoid loss (2/3 eyes, 66.7%). Time from initial to last SD-OCT varied from 1.0 to 56.0 weeks (mean 14.2; median 12.0). Persistent changes identified on SD-OCT at last visit included ONL thinning in 15 eyes (19.2%), ellipsoid zone defects partially or fully reversed in 8 eyes (10.3%), ellipsoid zone defects persisted in 7 eyes (9.0%), resolution of outer retinal hyper-reflectivity occurred in 4 eyes (5.1%), and hyporeflectivity of the photoreceptor/RPE complex remained in 1 eye.

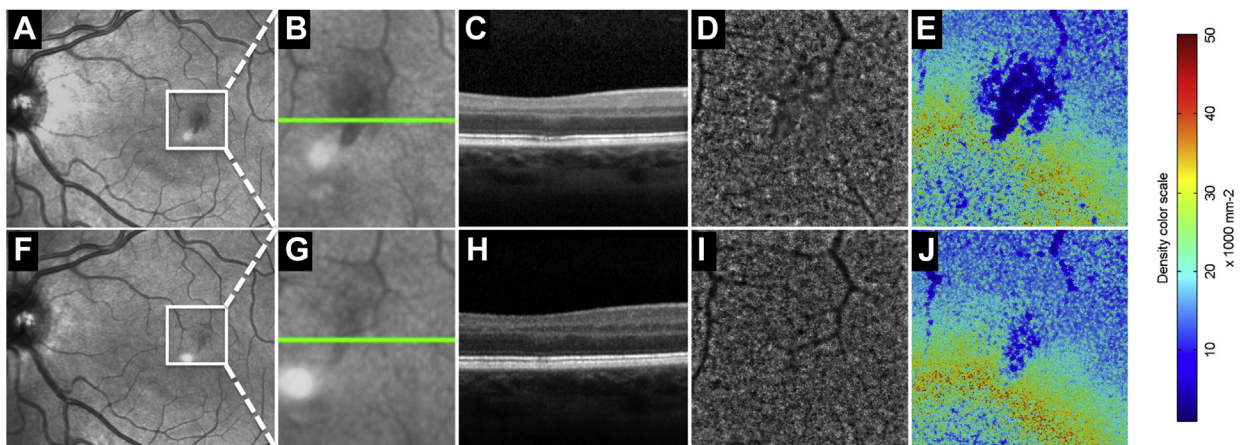


Fig. 4 – Multimodal imaging of acute macular neuroretinopathy (AMN) in a 21-year-old healthy woman presenting with acute-onset paracentral scotoma in the left eye. A, B: Near-infrared reflectance imaging revealed a teardrop-shaped dark gray lesion. C: Corresponding spectral-domain optical coherence tomography (SD-OCT) showed subtle hyper-reflectivity at the junction of the outer plexiform and outer nuclear layers, with disruption of the underlying ellipsoid and interdigitation zones. Adaptive optics revealed heterogeneous disruption of the D: cone photoreceptor mosaic with E: cone density reduction within an area that extended beyond both the limits of the NIR lesion and zone of ellipsoid loss on SD-OCT. The automated color mapping of the cone density is based on a color scale (demonstrated to the right of the map) expressed in thousands of cones/mm². After 5 months, the AMN lesion was barely detectable with F, G: NIR imaging, and there was near-complete resolution of the ellipsoid and interdigitation zones as demonstrated by H: SD-OCT. Follow-up adaptive optics revealed improvement, but incomplete recovery of the I: cone mosaic with J: persistent cone photoreceptor loss at the level of the lesion. (Images courtesy of Sarah Mrejen, MD.)

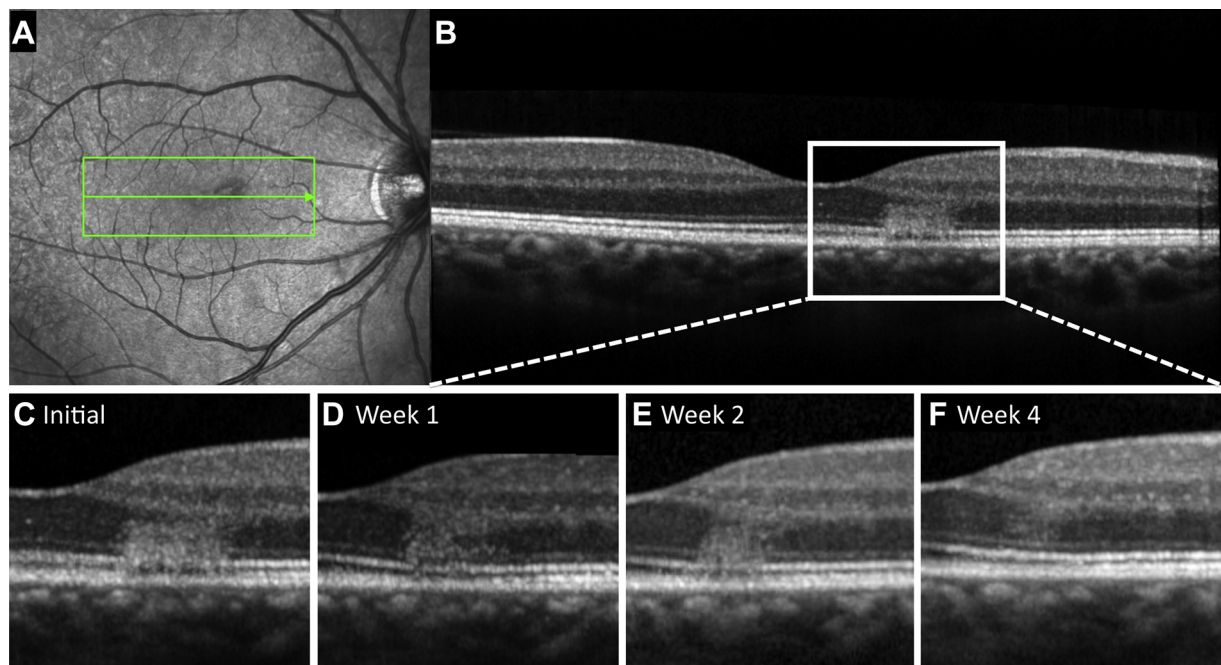


Fig. 5 – Improvement of acute macular neuroretinopathy demonstrated with serial spectral-domain optical coherence tomography (SD-OCT) scans in a 43-year-old healthy woman who had initially presented with a 4-day history of a “claw-shaped” scotoma just temporal to fixation in her right eye. A: Near-infrared reflectance imaging revealed a paracentral, well-demarcated, horseshoe-like dark lesion nasal to fixation in the right eye. B, C: Initial SD-OCT showed a hyper-reflective band involving the outer plexiform layer (OPL) and outer nuclear layer (ONL) with associated disruption of the ellipsoid and interdigitation zones. D–F: Serial SD-OCT scans obtained over the ensuing 4 weeks displayed gradual improvement of the hyper-reflectivity with persistence of punctate hyper-reflective areas within the OPL and ONL, whereas the ellipsoid and interdigitation zones were restored. (Reproduced from Sarraf et al⁴⁴ with permission from American Medical Association.)

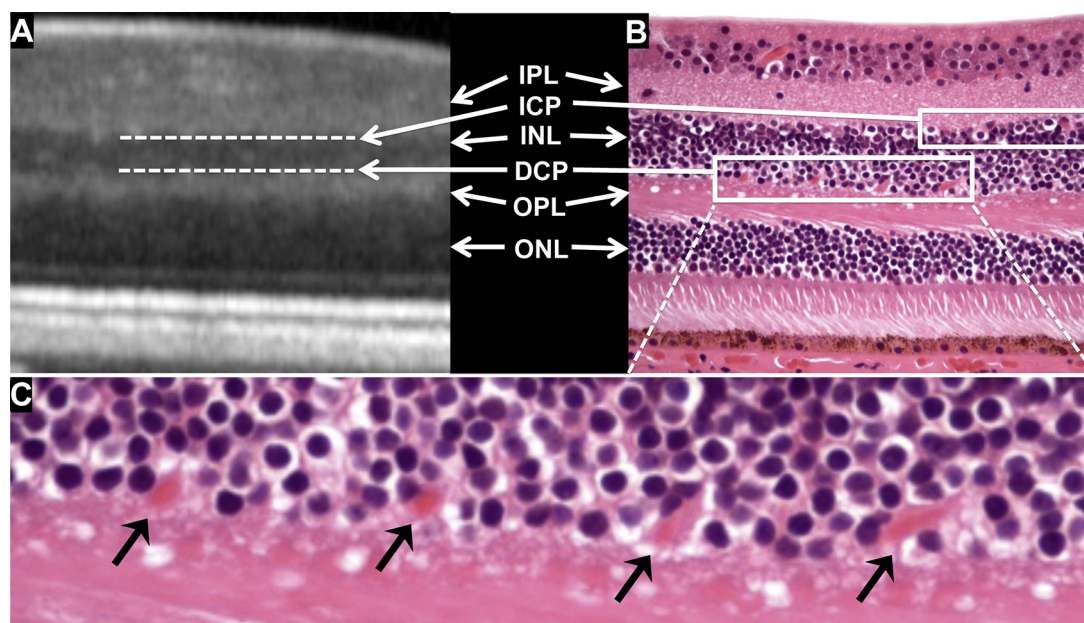


Fig. 6 – Histology of the intermediate capillary plexus (ICP) and deep capillary plexus (DCP). A: Spectral-domain optical coherence tomography image and B: histologic correlate of a normal macula demonstrating the location of the ICP (top interrupted line) and DCP (bottom interrupted line) along the apical and basal surfaces of the inner nuclear layer, respectively. C: Magnified histologic view of the microvascular network denoting individual vascular units of the DCP (arrows). (Reproduced from Sarraf et al⁴⁴ with permission from American Medical Association. Histology image courtesy of Ralph Eagle, MD.)

2.4.6. Electrophysiology

Results of electrophysiological testing were reported in 46 of 156 eyes (29.5%), including 28 eyes (17.9%) that underwent full-field electroretinography (ERG), 20 eyes (12.8%) that underwent multifocal electroretinography (mfERG), and 6 eyes (3.8%) that underwent pattern ERG. Among the eyes that underwent full-field ERG, 25 eyes (89.2%) showed no abnormalities, and 3 eyes (10.7%) showed one or more abnormal findings including reduced a-wave in 1 eye (3.6%), reduced a-wave and b-wave in 1 eye (3.6%), and reduced rod and cone response in 1 eye (3.6%). Of the eyes that underwent mfERG, only one (5.0%) showed no abnormalities and 19 eyes (95.0%) showed one or more abnormal findings, including diminished amplitudes in 19 eyes (95.0%) and diminished implicit time in 1 eye (5.0%). Among the eyes that underwent pattern ERG, 2 eyes (33.3%) showed no abnormalities and 4 eyes had diminished amplitudes (66.7%).

Follow-up electrophysiological testing was limited. Follow-up pattern ERG was available in 4 eyes (66.7%) of 2 patients and revealed nearly complete normalization of amplitudes and latencies at 3 days in 1 patient and 16 months in the other patient. Follow-up mfERG in 1 eye (5.0%) showed incomplete recovery of waveform amplitudes with persistent depression of the central macula at 3 months. No follow-up full-field ERG testing is reported.

2.4.7. Adaptive optics

Adaptive optics was reported in 6 of 156 eyes (3.8%). Mrejen and colleagues⁴⁵ described 5 eyes of 4 patients with reduced

cone photoreceptor density at the level of AMN lesions (Fig. 4). Furthermore, in these eyes, the cone mosaic disruption appeared heterogeneous and more widespread than the lesion detected using either NIR or SD-OCT imaging. There was incomplete recovery of the cone photoreceptor mosaic in 3 eyes at final follow-up (mean 4.7 months, range 3–6 months) despite complete resolution of the AMN lesion using NIR imaging.

In a study by Hansen and colleagues,²⁸ adaptive optics scanning light ophthalmoscope imaging of the photoreceptor mosaic revealed areas of both non-wave-guiding cones and decreased cone density. Follow-up at 6 months revealed no change in cone structure.

3. Discussion

Our comprehensive review of published literature suggests that AMN affects a demographically distinct group of patients. Over 80% of patients were women, and over half were in the third decade of life. Race was described in approximately one-third of reviewed cases, and over three-fourths of these subjects were reported to be non-Latino white.

Analysis of reported risk factors in AMN would seem to suggest a retinal vascular etiology. The most frequent associations were nonspecific illness in approximately half of the reviewed cases, followed by oral contraceptive use in over one-third of cases. Additional potentially vasoactive events included exposure to epinephrine or ephedrine, trauma,

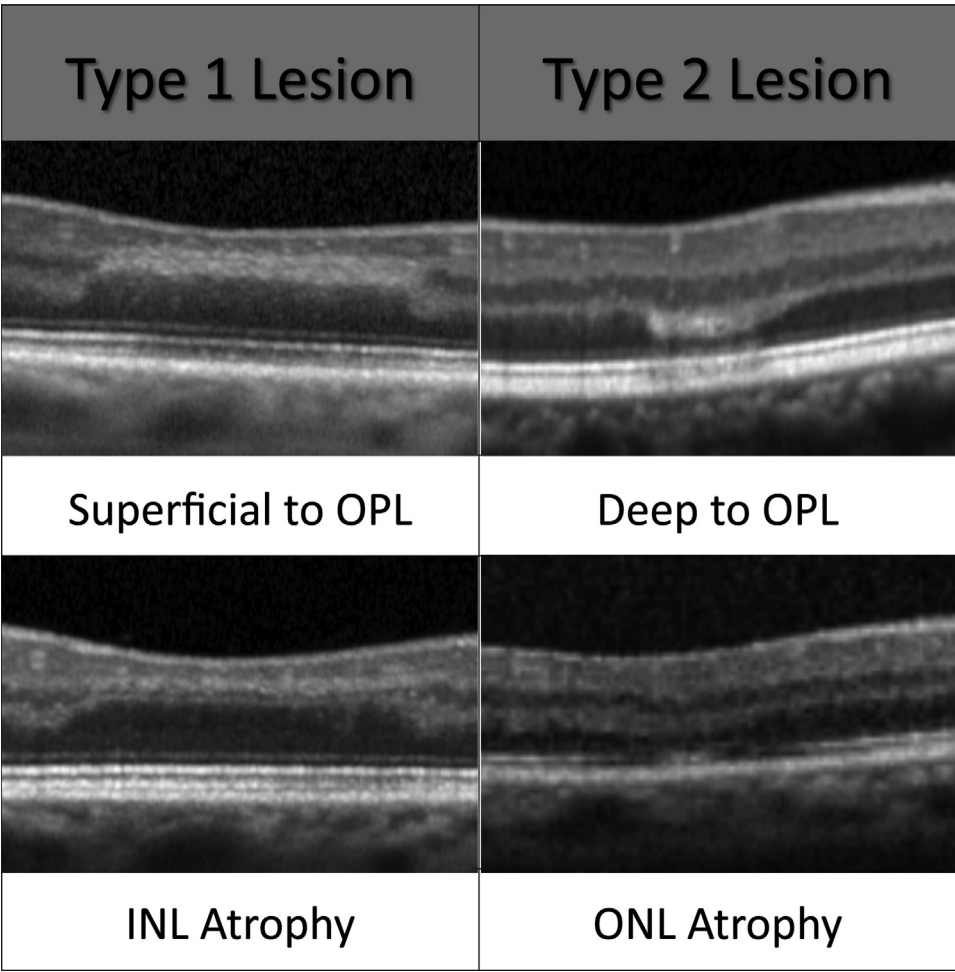


Fig. 7 – Proposed classification scheme of acute macular neuroretinopathy reflecting the location of spectral-domain optical coherence tomography lesions. Type 1 lesions, also known as paracentral acute middle maculopathy (PAMM), localize above the outer plexiform layer (OPL), with inner nuclear layer (INL) involvement. PAMM lesions eventually resolve with INL atrophy. Conversely, type 2 lesions localize below the OPL, with outer macular involvement. This type ultimately results in outer nuclear layer atrophy.

dehydration, hypovolemia, and systemic shock. Additional possible associations such as dengue fever, anemia, ulcerative colitis, thrombocytopenia, lupus, and leukemia have been only recently suggested in a few patients and were not included in our analysis.^{32,35,37,47}

In 2015, Monk and colleagues⁴⁷ reported several previously unrecognized potential etiologies of classic AMN, including dengue fever, anemia, ulcerative colitis, thrombocytopenia, leukemia, Valsalva maneuver, as well as the sympathomimetic agent Lis-dexamphetamine dimesylate. In addition to previously implicated mechanisms of dehydration and hypovolemia, they also propose hyperviscosity from leukocytosis, increased capillary permeability, endothelial dysfunction and hemorrhagic diathesis, platelet destruction, occlusion of precapillary arterioles by immune complex deposition, and consumptive coagulopathy as potential etiologies of focal deep capillary ischemia. In addition, in 2015, Li and colleagues³⁷ described 9 eyes of 5 patients that exhibited AMN concomitantly with classic dengue maculopathy, Introini and colleagues³² linked AMN with intranasal cocaine

use, and Lee and colleagues³⁵ reported the first case of AMN associated with SLE: all bolstering the growing evidence of an underlying vascular process in AMN.

Review of results obtained with various imaging techniques highlighted findings characteristic of AMN. In one-third of cases, AMN lesions appeared hyporeflective with well-demarcated margins using NIR and were located eccentrically within the perifoveal region. The hyporeflective abnormalities detected on NIR imaging correlated with ellipsoid and interdigitation zone (IZ) disruption on SD-OCT and the latter may represent a subclinical alteration in RPE melanin.¹⁸ Normalization of the IZ was correspondingly associated with fading of the hyporeflective NIR lesions. Near-infrared imaging in conjunction with SD-OCT may be the most sensitive imaging modality to diagnose AMN and may be abnormal even when clinical examination and color fundus photography fail to detect an abnormality. As with AMN, the increased penetration of NIR may be diagnostically useful in other diseases that primarily affect the outer retina.

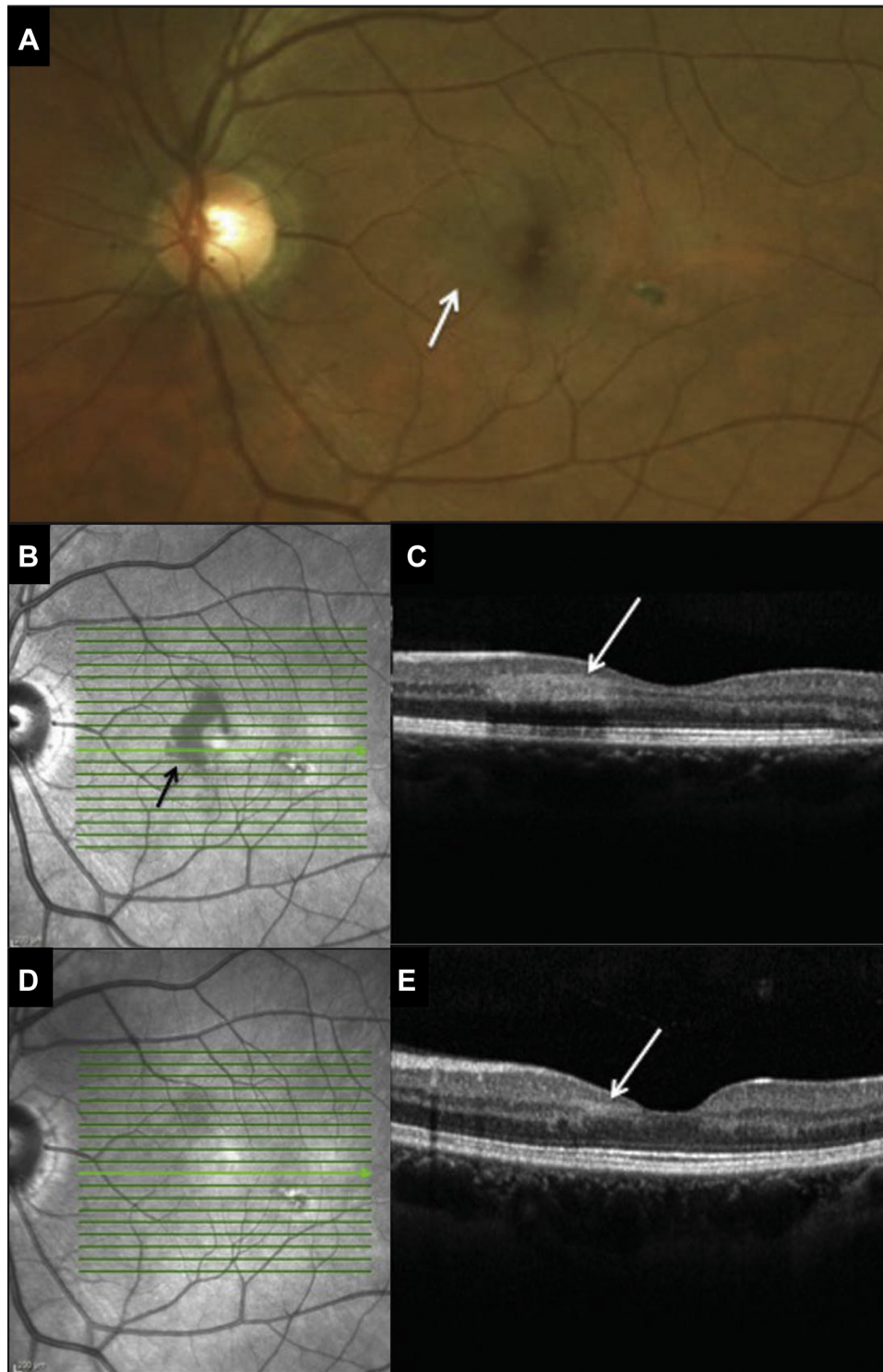


Fig. 8 – Paracentral acute middle maculopathy (PAMM) in a 32-year-old healthy woman who reported drinking several cups of coffee daily and presented with a 1-week history of a paracentral scotoma in the left eye. A: Visual acuity was 20/20 at presentation, and retinal examination of the left eye demonstrated a subtle, gray-white, wedge-shaped lesion (*arrow*), and an unrelated small chorioretinal scar temporally. B: Near-infrared reflectance (NIR) imaging delineated the paracentral, dark gray lesion affecting the nasal fovea of the left eye (*arrow*). C: Corresponding spectral-domain optical coherence tomography imaging revealed a characteristic hyper-reflective, band-like plaque at the level of the inner nuclear layer (INL) (*arrow*), consistent with PAMM and type 1 AMN. D: Follow-up imaging after 3 months revealed resolution of the dark gray paracentral lesion on NIR, and E: subsequent thinning of the INL by SD-OCT (*arrow*). (Reproduced from Tsui and Sarraf,⁵¹ with permission from SLACK Incorporated.)

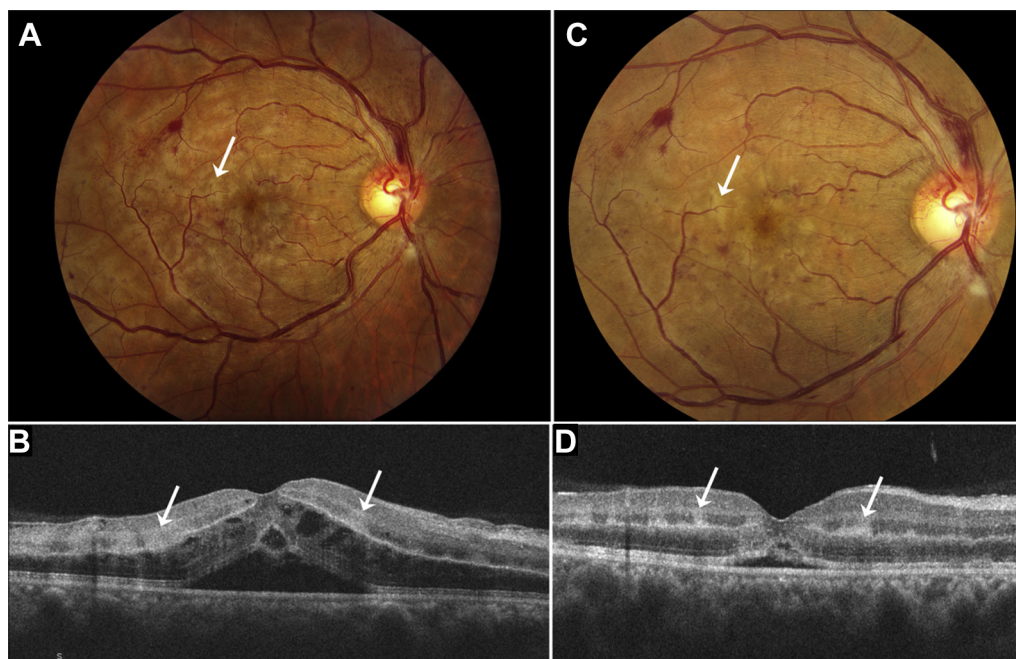


Fig. 9 – Paracentral acute middle maculopathy (PAMM) in the setting of nonischemic central retinal vein occlusion (CRVO). A 32-year-old healthy man with recent upper respiratory infection presented with a 6-day history of acute vision loss in the right eye. **A:** Visual acuity was 5/400 at presentation, and retinal examination of the right eye was consistent with acute CRVO and associated deep retinal whitening in a perivenular distribution in the temporal macula (white arrow). **B:** High-definition spectral-domain optical coherence tomography (SD-OCT) revealed macular edema with intraretinal and subretinal fluid and multiple hyper-reflective, plaque-like lesions involving the inner nuclear layer (white arrows), consistent with PAMM and ischemia of the DCP. At 1-week follow-up, the lesions on **C:** fundusoscopic examination and **D:** SD-OCT were still present (white arrows); however, the subretinal fluid had markedly improved, accounting for an improvement in visual acuity to 20/250 in the affected eye. (Reproduced from Rahimy and Sarraf,³⁷ with permission from Wolters Kluwer Health.)

Fluorescein and ICG angiography were largely unremarkable in most cases of AMN. Nonetheless, these imaging techniques may be helpful in excluding other causes of acute vision loss, including white dot syndromes such as acute posterior multifocal placoid pigment epitheliopathy and multiple evanescent white dot syndrome, which also tend to occur in young patients and have characteristic angiographic findings. As a result of the shared demographic, AMN has traditionally and incorrectly been grouped with the white dot syndromes. It is now apparent that AMN is largely the result of noninflammatory ischemia of retinal capillaries, likely of the deep capillary plexus.⁵⁴

FAF imaging was rarely reported ($n = 8$ cases) and was largely unremarkable. Subtle hyperautofluorescence corresponding to clinically observed lesions was rarely observed. Normal FAF suggests a lack of involvement of the RPE in AMN and can be used to exclude other conditions associated with acute vision loss, such as acute idiopathic maculopathy or the aforementioned white dot syndromes such as multiple evanescent white dot syndrome and acute posterior multifocal placoid pigment epitheliopathy.³³

SD-OCT was exceedingly sensitive in detecting structural abnormalities associated with AMN. Nearly all reported eyes showed one or more abnormal findings, including ellipsoid zone disruption in half of the imaged eyes and

hyper-reflectivity of the outer nuclear layer in one-third of imaged eyes. More recently, early hyper-reflective plaques at the border of the OPL and ONL have been identified and may be a particularly early and characteristic sign of AMN.^{4,5,18} These hyper-reflective plaques typically progress to thinning of the ONL over time.

SD-OCT was also valuable in following structural changes over time and revealed that such changes persisted in fewer than one-third of affected eyes. Fawzi and colleagues¹⁸ described a case of ONL thinning, attenuation of the ellipsoid zone, and persistent absence of the interdigitation zone (or IZ, the junction between the outer segments and retinal pigment epithelium) in a patient at 14-month follow-up. Near-complete resolution of outer retinal SD-OCT abnormalities has been documented to occur as early as 4 weeks (Fig. 5).^{40,44} Vance and colleagues⁷¹ noted focal loss of the ellipsoid zone accompanied by subtle ONL thinning in the acute period. Subsequently, the ellipsoid zone was re-established at 1-month follow-up, but persistent ONL thinning remained.

AMN was originally described as an inner retinal disease on the basis of clinically observed lesions. The etiology is likely complex and remains to be firmly established. Infection, inflammation, and ischemia have all been postulated in the pathogenesis of this condition. Turbeville and colleagues⁶⁹

suggested that a vascular etiology may unify these disparate associations.

The advent of sophisticated retinal imaging devices such as SD-OCT has enabled better characterization of the lesions in AMN and has enhanced our understanding of the relevant pathophysiologic mechanisms including vascular compromise or ischemia. Localization of the lesions to the outer macula including defects of the ellipsoid zone and IZ in association with abnormalities of the OPL and ONL have been described by multimodal imaging, implicating ischemia of deep retinal capillary plexus.¹⁸ Prior animal studies demonstrated the presence of 4 distinct capillary plexuses.^{54,66} Most superficial is the radial peripapillary network that supplies the dense nerve fiber layers around the disk and is in close approximation to the more proximal portions of the arcade vessels. Just beneath the radial peripapillary network and extending throughout the macula is the superficial capillary network, which supplies the ganglion cell layer. Below the superficial capillary network are both the intermediate and deep capillary plexuses that straddle the inner nuclear layer (Fig. 6).⁵⁴ As AMN lesions typically develop at the junction between the OPL and ONL and are frequently associated with disruption of the both the EZ and IZ, it seems most probable that these changes result from local compromise of the deep capillary plexus which provides retinal perfusion to the vulnerable watershed zone between the retinal and choroidal circulations. This concept of deep capillary plexus ischemia causing AMN is strongly supported by previously identified risk factors for the condition, including nonspecific flu-like illness or fever, use of oral contraceptives, exposure to either epinephrine or ephedrine, antecedent trauma, and systemic shock—each of which could directly or indirectly influence retinal capillary perfusion.

Paracentral acute middle maculopathy (PAMM) was recently recognized as the development of band or plaque-like hyper-reflective SD-OCT lesions at the level of the INL which are similar to, but more superficial than, the hyper-reflective SD-OCT bands seen in patients with AMN.^{52,53,55,61,75} Although Sarraf and colleagues⁶¹ initially identified 2 SD-OCT presentations of AMN (Figs. 7 and 8), it is now clear that PAMM and AMN are distinct disorders with some overlapping features. The formation of PAMM lesions in the setting of nonproliferative diabetic retinopathy, retinal vein and arterial occlusion, sickle cell disease, Purtscher retinopathy, and retinal vasculitis further strengthens the idea that vascular disease (local and systemic) can cause isolated deep retinal capillary ischemia and provides indirect support that AMN may also result from retinal microvascular insufficiency (Fig. 9).^{8,11,52,53,55,61,68,75,76} Each entity tends to cause eccentric wedge-shaped lesions that extend to the edge of the foveal avascular zone and are associated with corresponding scotomas that are often permanent; however, PAMM lesions are defined by the SD-OCT presence of a hyper-reflective plaque at the level of the INL, which is straddled by intermediate and deep retinal capillary plexuses. Permanent scotomas in these patients may be related to permanent thinning of the INL that ensues after the initial hyper-reflective infarct. In contrast, AMN demonstrates hyper-reflective plaques at the level of the OPL/ONL junction with associated disruption of the EZ and IZ. Outer nuclear

layer thinning may ensue and may also lead to permanent paracentral scotomas. The presence of associated paracentral ellipsoid loss may be explained on the basis of oxygen supply to the photoreceptors that is partially contributed (10%) by the deep retinal capillary plexus.

Conventional fluorescein angiography appears to be inadequate in defining the morphology of the intermediate and deep retinal capillary plexuses in human subjects. Animal studies have suggested that scattering from the retinal layers may compromise the image resolution of the deeper capillary systems. The recent advent of depth-resolved en face optical coherence tomography angiography (OCTA) offers the ability to segment each of the retinal capillary networks and further characterizes the retinal microvasculature and its potential pathology in cases of PAMM and AMN.^{10,54}

4. Summary

AMN preferentially affects young, non-Latino white women and is most commonly associated with nonspecific flu-like illness, fever, or oral contraceptive use. Multimodal diagnostic imaging has enabled an increasingly detailed characterization of AMN, and SD-OCT has provided anatomical data to further support the critical role of deep capillary plexus compromise. The utilization of segmentation software and OCTA may allow for enhanced visualization of the intermediate and deep capillary plexuses. Ultrahigh resolution OCT in conjunction with adaptive optics may be used to define the retinal capillary morphology in 3 dimensional quality.⁷² Animal studies to support causality and histopathological correlation that are currently lacking may, in the end, be the most important modality to prove definitively a microvascular mechanism.

5. Method of literature search

This updated review includes all reports of AMN in the English and non-English language peer-reviewed literature published from the first report by Bos and Deutman⁴ in 1975 through December 2014. PUBMED and MEDLINE databases were searched using the keywords acute macular neuroretinopathy, AMN, AMNR, AMOR, multimodal imaging, and optical coherence tomography. Cited studies not identified in PUBMED and MEDLINE were located by reviewing the references detected in the aforementioned search or by reviewing reference lists in relevant textbook chapters.

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